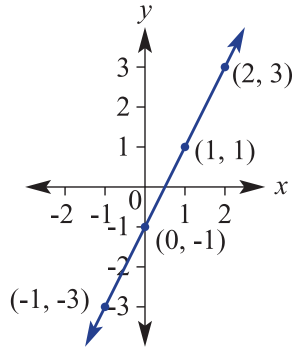
# 5.7 Gradient–intercept form

Shown here is the graph of the rule $y = 2x - 1$. It shows a gradient of 2 and a $y$-intercept of -1. The fact that these two numbers match numbers in the rule is no coincidence. This is why rules written in this form are called gradient–intercept form. Other examples of rules in this form include:

$$y = -5x + 2,\ y = \frac{1}{2}x - 0.5 \text{ and } y = \frac{x}{5} + 20.$$




## Let's start: What's in common?

Sketch the following linear relations on the same set of axes. Plot points or use the $x$- and $y$-intercepts.

**a** $y = 2x$ **b** $y = 2x + 2$ **c** $y = 2x - 1$

- What do these graphs have in common?
- Calculate the gradient of each line. What do you notice?
- How do the gradients relate to the rules?

Now, sketch the following linear relations on the same set of axes. Plot points or use the $x$- and $y$-intercepts.

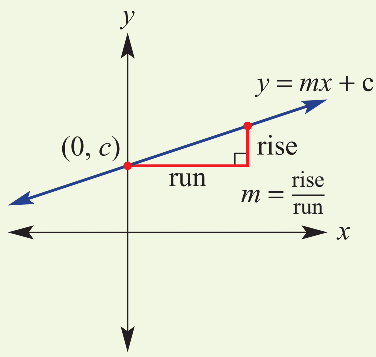
**a** $y = x + 1$ **b** $y = 2x + 1$ **c** $y = -\frac{1}{2}x + 1$

- What do these graphs have in common?
- How does the common feature relate to the rules?
- Can you form a rule for a linear relation with a gradient of 3 and a $y$-intercept at 2?

## Key ideas

$m$ = gradient $c$ = $y$-intercept

- $y = mx + c$ (or $y = mx + b$ depending on preference) is the **gradient–intercept form** of a straight line equation.
- If the $y$-intercept is zero, the equation becomes $y = mx$ and these graphs will pass through the origin.
- Any linear relation can be rearranged to be written in gradient–intercept form by making $y$ the subject.
- To sketch a graph using the gradient–intercept method, find the $y$-intercept and use the gradient to find a second point.
  - For example, if $m = \frac{2}{5}$, move 5 across and 2 up from the $y$-intercept.
  - For the gradient of $-2 = \frac{-2}{1}$ you would move 1 across and 2 down from the $y$-intercept.




**Gradient–intercept form** The equation of a straight line, written with $y$ as the subject of the equation

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# Exercise 5G

**Understanding**

**1** In the gradient–intercept form $y = mx + c$:

- **a** $m$ is the ________.
- **b** $c$ is the ________.
- **c** $y$ is the ________ of the equation.

**2** Match the given gradient and $y$-intercept in the first column with the rule in the left column.

| | |
|---|---|
| **a** gradient = 2, $y$-intercept = 5 | **i** $y = 5x + 2$ |
| **b** gradient = 5, $y$-intercept = 2 | **ii** $y = -2x + 3$ |
| **c** gradient = -2, $y$-intercept = 3 | **iii** $y = 2x + 5$ |
| **d** gradient = -1, $y$-intercept = -2 | **iv** $y = -x - 2$ |

**3** **a** From the point (0, 2), give the coordinates of the point that is:

**i** 2 across and 4 up **ii** 3 across and 1 down **iii** 4 across and 6 down

**b** From the point (0, -1), give the coordinates of the point that is:

**i** 1 across and 2 up **ii** 3 across and 4 down

Plot the point to help you visualise, if required.

**4** Fill in the missing numbers.

- **a** The gradient $\frac{5}{3}$ describes a run of _____ for a rise of _____.
- **b** The gradient 4 describes a run of _____ for a rise of _____.
- **c** The gradient $\frac{-1}{2}$ describes a run of _____ for a rise of _____.

Gradient = $\frac{\text{rise}}{\text{run}}$.

**Fluency**

## Example 18 Stating the gradient and $y$-intercept

State the gradient and the $y$-intercept for the graphs of the following relations.

**a** $y = 2x + 1$ **b** $y = -3x$

| Solution | Explanation |
|---|---|
| **a** $y = 2x + 1$<br>The gradient = 2<br>$y$-intercept = 1 | The rule is given in gradient intercept form, $y = mx + c$.<br>The gradient is the coefficient of $x$ (the numeral multiplied by $x$).<br>The constant term is the $y$-intercept. |
| **b** $y = -3x$<br>The gradient = -3<br>$y$-intercept = 0 | The gradient is the coefficient of $x$ including the negative sign.<br>The constant term is not present so the $y$-intercept = 0. |

**5** State the gradient and $y$-intercept for the graphs of the following relations.

**a** $y = 3x - 4$ **b** $y = -5x - 2$ **c** $y = -2x + 3$ **d** $y = \frac{1}{3}x + 4$

**e** $y = -4x$ **f** $y = 2x$ **g** $y = 2.3x$ **h** $y = -0.7x$

$$\underset{\text{gradient}}{y = m}x + \underset{y\text{-intercept}}{c}$$

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## Example 19 Rearranging linear equations into the form $y = mx + c$

Rearrange these linear equations into the form $y = mx + c$.

**a** $2x + y = 7$ **b** $4x + 2y = 10$

| Solution | Explanation |
|---|---|
| **a** $2x + y = 7$<br>$y = 7 - 2x$<br>$y = -2x + 7$ | To have $y$ by itself, subtract $2x$ from both sides.<br>$7 - 2x$ is the same as $-2x + 7$, which is in the form $mx + c$. |
| **b** $4x + 2y = 10$<br>$2y = -4x + 10$<br>$y = -2x + 5$ | Solve for $y$ by first subtracting $4x$ from both sides.<br>Divide both sides of the equation by 2:<br>$\frac{-4x+10}{2} = \frac{-4}{2}x + \frac{10}{2}$. |

**6** Rearrange these linear equations into the form $y = mx + c$.

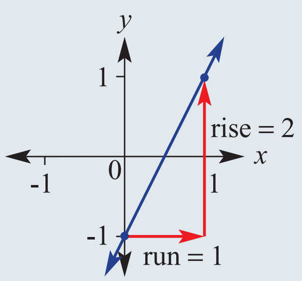
**a** $3x + y = 4$ **b** $2x + y = 7$ **c** $y - 3x = 2$
**d** $y - 2x = 7$ **e** $5x - y = 3$ **f** $4x + 2y = 8$
**g** $3x + 2y = 6$ **h** $4x - 2y = 5$ **i** $5x - 3y = -6$

Follow steps as if you were solving an equation for $y$.

## Example 20 Sketching linear graphs using the gradient and $y$-intercept

Find the value of the gradient and $y$-intercept for these relations and sketch their graphs.

**a** $y = 2x - 1$ **b** $y = \frac{-3}{4}x + 2$

| Solution | Explanation |
|---|---|
| **a** $y = 2x - 1$<br>$y$ intercept $= -1$<br>gradient $= 2 = \frac{2}{1}$ | The rule is in gradient–intercept form so we can read off the gradient (coefficient of $x$) and the $y$-intercept (constant term).<br>Gradient $= \frac{2 \text{ (rise)}}{1 \text{ (run)}}$; for every 1 across, move 2 up. |
|  | Label the $y$-intercept at (0, -1).<br>From (0, -1) move 1 across (to the right) and 2 up to give a second point at (1, 1).<br>Mark and join the points to form a line. |

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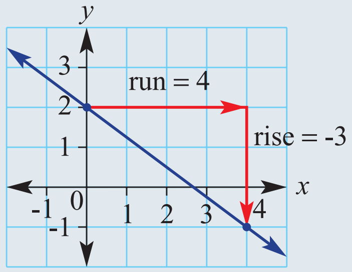
**b** $y = \frac{-3}{4}x + 2$

$y$-intercept $= 2$

gradient $= \frac{-3}{4}$




Read off the $y$-intercept and gradient from the gradient–intercept form.
Rise = -3, run = 4.

Label the $y$-intercept (0, 2) and from here move 4 across to the right (run) and 3 down (due to -3) to give a second point at (4, -1).
Join points to form a line.

**7** Find the gradient and $y$-intercept for these relations and sketch their graphs.

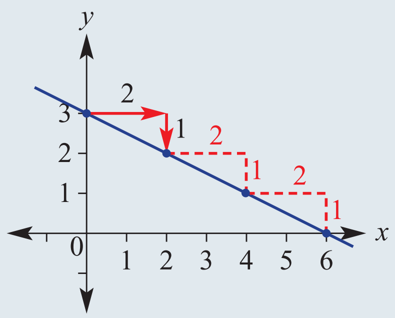
**a** $y = x - 2$ **b** $y = 2x + 1$ **c** $y = 3x + 2$ **d** $y = \frac{1}{2}x + 2$

**e** $y = -3x + 3$ **f** $y = \frac{3}{2}x + 1$ **g** $y = 2x$ **h** $y = \frac{-4}{3}x$

Plot the $y$-intercept first, then use the rise and run of the gradient.

## Example 21 Rearranging equations to sketch using the gradient and $y$-intercept

Rearrange the equation $x + 2y = 6$ to find the gradient and $y$-intercept, and sketch its graph.

| **Solution** | **Explanation** |
|---|---|
| $x + 2y = 6$<br>$2y = 6 - x$<br>$y = 3 - \frac{1}{2}x$<br>$= -\frac{1}{2}x + 3$<br>so $y$-intercept is 3 | Make $y$ the subject by subtracting $x$ from both sides and then dividing both sides by 2.<br>$\frac{6-x}{2} = \frac{6}{2} - \frac{x}{2} = 3 - \frac{1}{2}x$<br>Rewrite in the form $y = mx + c$ to read off the gradient and $y$-intercept. |
| gradient $= -\frac{1}{2} = \frac{-1}{2}$ | Link the negative sign to the rise (-1) so the run is positive (+2). |
|  | Mark the $y$-intercept (0, 3) then from this point move 2 right and 1 down to give a second point at (2, 2).<br>Note that the $x$-intercept will be 6. If the gradient is $\frac{-1}{2}$ then a fall of 3 needs a run of 6. |



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**8** Rearrange these equations to find the gradient and $y$-intercept and sketch their graphs.

**a** $x + y = 4$ **b** $y - x = 6$ **c** $4x + 2y = 6$
**d** $x + 2y = 8$ **e** $2x + 3y = 6$ **f** $4x + 3y = 12$
**g** $x - 2y = 4$ **h** $2x - 3y = 6$ **i** $3x - 4y = 12$
**j** $x + 4y = 0$ **k** $x - 5y = 0$ **l** $8x - 2y = 0$

Solve for $y$ to make $y$ the subject with equations in the form $y = mx + c$.

**Problem-solving and Reasoning**

**9** During a heavy rainstorm, a rain gauge is filling with water. The rule for the volume of water in the tank ($V$ mL), $t$ hours after the start of the storm, is given by $V = 4t + 2$.

**a** State the gradient and $V$-intercept of this relation and sketch its graph.
**b** In this scenario, what do the gradient and the $V$-intercept represent?

Here, the $V$-intercept is like the $y$-intercept.


**10** Vera is trying to convince her friend Li that $\dfrac{4x+6}{2} = 2x + 6$. She offers to pay Li \$2 if she is wrong. Does Vera lose her money? Explain.

**11** Which of these linear relations have a gradient of 2 and $y$-intercept of -3?

**a** $y = 3 - 2x$ **b** $y = \dfrac{2x-6}{2}$ **c** $y = \dfrac{4x-6}{2}$
**d** $y = \dfrac{3-2x}{-1}$ **e** $2y = 4x - 3$ **f** $-2y = 6 - 4x$

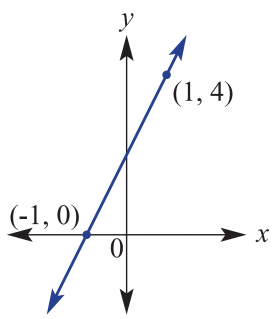
Rewrite each in the form $y = mx + c$ and look for $y = 2x - 3$.

**12** Jeremy says that the graph of the rule $y = 2(x + 1)$ has gradient 2 and $y$-intercept 1.

**a** Explain his error.
**b** What can be done to the rule to help show the $y$-intercept?

## The missing $y$-intercept

**13** This graph shows two points (-1, 0) and (1, 4), with a gradient of 2. By considering the gradient (2 up for every 1 across), the $y$-intercept can be calculated to be 2, so $y = 2x + 2$.




Use this approach to find the rule of the line passing through these points.

**a** (-1, 1) and (1, 5) **b** (-2, 4) and (2, 0)
**c** (-2, -1) and (3, 9) **d** (-4, 1) and (2, 4)
**e** (-1, 3) and (1, 4)

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# 5.8 Finding the equation of a line

When data points from an experiment are plotted, they may form a straight line. This shows a linear relationship between the two variables, such as time and growth, involved in the experiment.

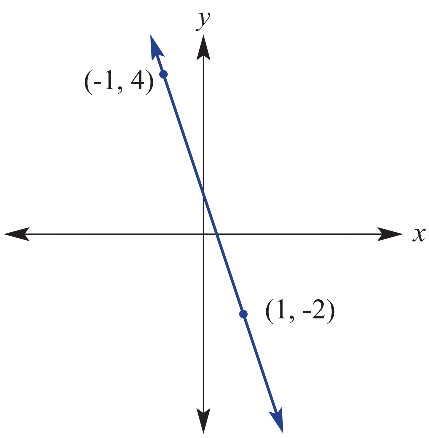
By finding the equation of this line, we can develop a rule to relate the two variables. This rule can be used to make further predictions about the data, and could be applied to a larger scale experiment.

Using gradient–intercept form, the rule (or equation) of a line can be found by calculating the value of the gradient and the $y$-intercept.

## Let's start: But we don't know the $y$-intercept!

A line with the rule $y = mx + c$ passes through two points (-1, 4) and (1, -2).

- Using the information given, is it possible to find the value of $m$? If so, calculate its value.
- The $y$-intercept is not given on the graph. Discuss what information could be used to find the value of the constant, $c$, in the rule. Is there more than one way you can find the $y$-intercept?
- Write the rule for the line.




## Key ideas

- To find the equation of a line in gradient–intercept form, $y = mx + c$, you need to find:
  - the value of the gradient ($m$) using $m = \frac{\text{rise}}{\text{run}}$
  - the value of the constant ($c$), by observing the $y$-intercept or by substituting another point.
- All the points $(x, y)$ on a line satisfy the equation of the line.
  For example, (2, 5) is on the line with equation $y = 2x + 1$ since substituting $x = 2$ and $y = 5$ gives a true equation: $5 = 2 \times (2) + 1$.

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# Exercise 5H

Understanding

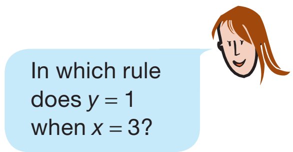
**1** Fill in the missing words.
To find the equation of a line in the form $y = mx + c$, the ________ ($m$) is required as well as the ________ ($c$) or another ________.

**2** Substitute the given values of $m$ and $c$ into $y = mx + c$ to write the rule.

**a** $m = 2, c = 5$ **b** $m = 4, c = -1$
**c** $m = -2, c = 5$ **d** $m = -1, c = \frac{-1}{2}$

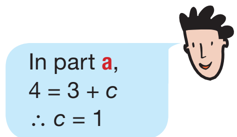
**3** Which one of the following lines does the point (3, 1) lie on?

**A** $y = 3x + 1$ **B** $y = x - 3$ **C** $y = -2x + 7$
**D** $y = 2x + 1$ **E** $y = -x - 2$




**4** Substitute the point into the given rule and solve to find the value of $c$.
For example, using (3, 4), substitute $x = 3$ and $y = 4$ into the rule.

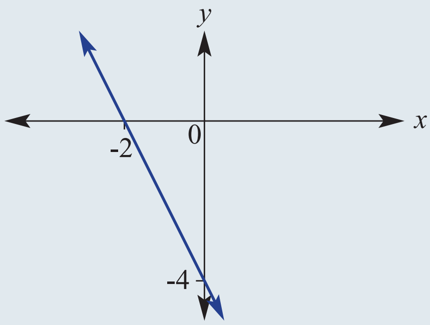
**a** $(3, 4), y = x + c$ **b** $(1, 5), y = 2x + c$ **c** $(-2, 3), y = 3x + c$
**d** $(-1, 6), y = 4x + c$ **e** $(3, -1), y = -2x + c$ **f** $(-2, 4), y = -x + c$



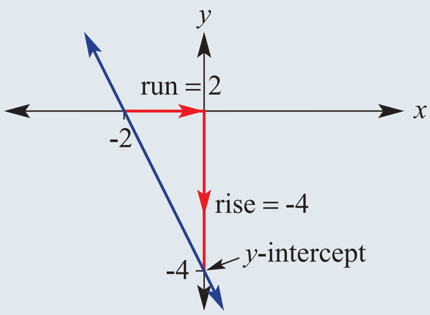

Fluency

## Example 22 Finding the equation of a line given the $y$-intercept and another point

Determine the equation of the straight line shown here.




| Solution | Explanation |
|---|---|
| The equation will have the form $y = mx + c$. | For the gradient–intercept form you need to find the gradient ($m$) and the $y$-intercept ($c$). |
| gradient, $m = \frac{\text{rise}}{\text{run}}$ $= \frac{-4}{2}$ $= -2$ | |
| $y$-intercept $= -4$. | |
| $\therefore y = -2x - 4$ | |




The $y$-intercept (-4) can be read from the graph.
In $y = mx + c$, substitute $m = -2$ and $c = -4$.

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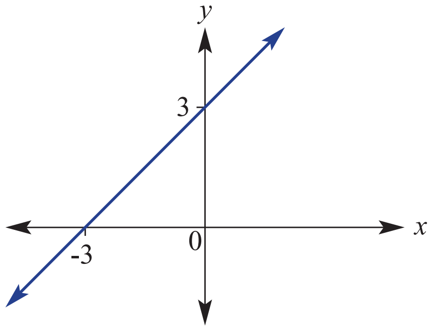
5H

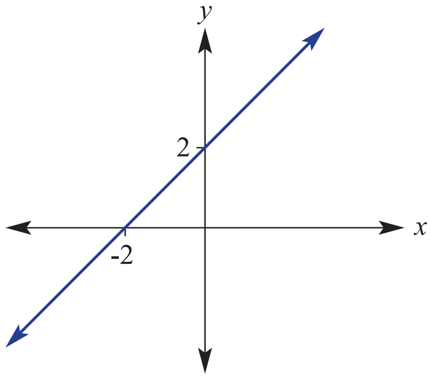
5 Determine the equation of the following straight lines.

Write in the form $y = mx + c$. You will need the gradient and the $y$-intercept (or another point if the $y$-intercept is not given).

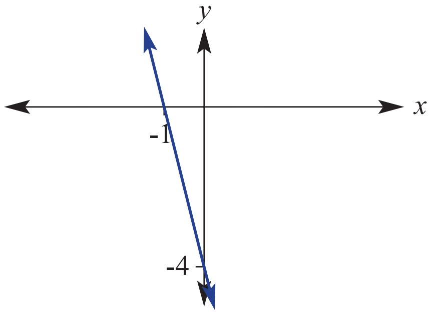
a



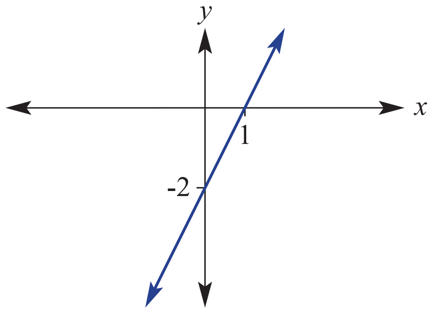

b



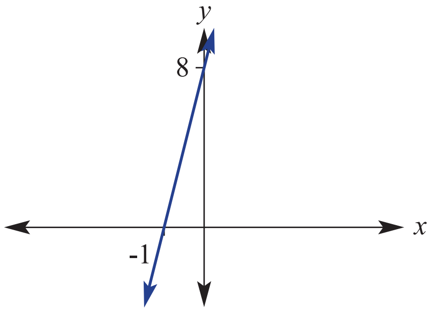

c



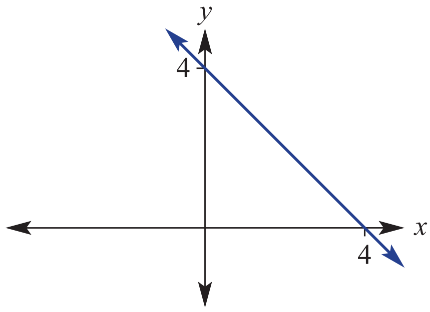

d



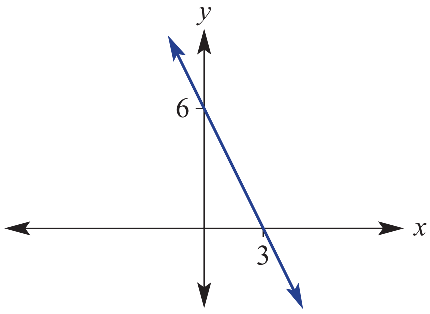

e




f




g

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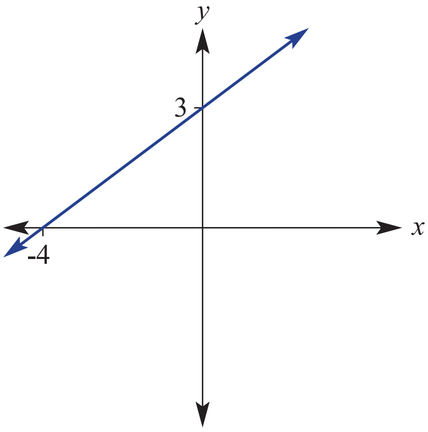
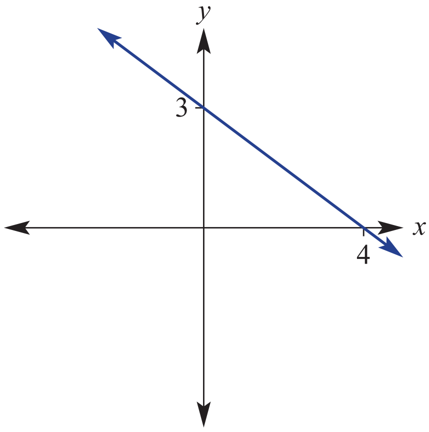
**6** Find the equation of these straight lines that have fractional gradients. Simplify any fractions.

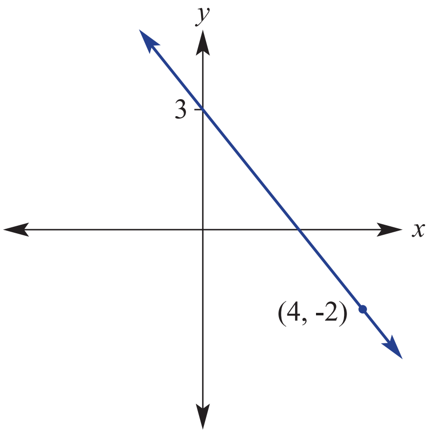
**a**



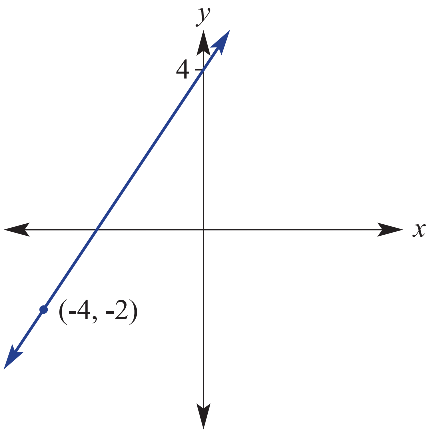

**b**



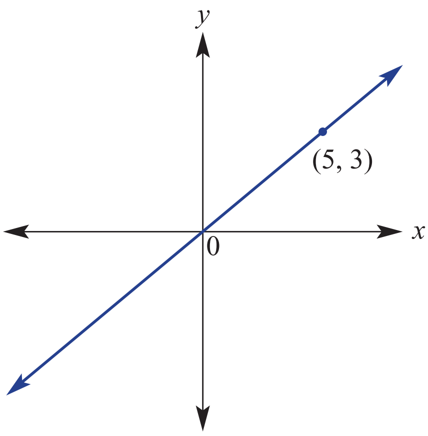

**c**



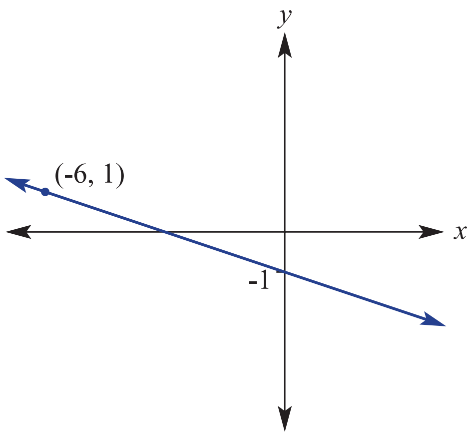

**d**




**e**




**f**

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5H

## Example 23 Finding the equation of a line given the gradient and a point

Find the equation of the line that has a gradient, $m$, of 3 and passes through the point (2, -1).

| Solution | Explanation |
|---|---|
| $y = mx + c$<br>$y = 3x + c$ | Substitute $m = 3$ into $y = mx + c$. |
| $-1 = 3 \times (2) + c$<br>$-1 = 6 + c$<br>$-7 = c$ | Since (2, -1) is on the line, it must satisfy the equation $y = 3x + c$, hence substitute the point (2, -1) where $x = 2$ and $y = -1$ to find $c$. Simplify and solve for $c$ by subtracting 6 from both sides. |
| $\therefore\ y = 3x - 7$ | Write the equation in the form $y = mx + c$. |

**7** Find the equation of the line that:

**a** has a gradient of 3 and passes through the point (1, 8)
**b** has a gradient of -2 and passes through the point (2, -5)
**c** has a gradient of -3 and passes through the point (2, 2)
**d** has a gradient of 1 and passes through the point (1, -2)
**e** has a gradient of -3 and passes through the point (-1, 6)
**f** has a gradient of 5 and passes through the point (2, 9)
**g** has a gradient of -1 and passes through the point (4, 4)
**h** has a gradient of -3 and passes through the point (3, -3)
**i** has a gradient of -2 and passes through the point (-1, 4)
**j** has a gradient of -4 and passes through the point (-2, -1)

In $y = mx + c$, insert the gradient value for $m$, then substitute the point for $x$ and $y$ to find $c$.

**Problem-solving and Reasoning**

**8** The coordinates (0, 1) mark the take-off point 1 metre above the ground for a rocket constructed as part of a science class. The positive $x$ direction is considered to be East. Find the equation of the rocket's path if it rises at a rate of 5 m vertically for every 1 m in an easterly direction.

**9** A line has gradient -2 and $y$-intercept 5. Find its $x$-intercept.

First write the rule of the line.

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